

Q3. (Filling Missing Lines) [5 Marks]

Complete the missing lines in the pseudocode below. The algorithm finds all factors of a given number and determines if it is a prime number. Fill in all six blanks (____).text

```
BEGIN
INPUT n
SET count = 0
OUTPUT "Factors:"
FOR i = 1 TO ____
IF n MOD i == ____ THEN
OUTPUT i
count = ____
ENDIF
ENDFOR
IF ____ == 2 THEN
OUTPUT n, "is prime"
ELSE
OUTPUT n, "is ____ prime"
ENDIF
OUTPUT "Total factors:", ____
END
```

Completed Pseudocode

```
BEGIN
INPUT n
SET count = 0
OUTPUT "Factors:"
FOR i = 1 TO n
IF n MOD i == 0 THEN
OUTPUT i
count = count + 1
ENDIF
ENDFOR
IF count == 2 THEN
OUTPUT n, "is prime"
ELSE
OUTPUT n, "is not prime"
ENDIF
OUTPUT "Total factors:", count
```

END